

A small library on Kolmogorov complexity in Basic Recursive Arithmetic

June 8, 2026

Abstract

This note records, in ordinary mathematical notation, the results about Kolmogorov complexity that have been machine-checked in the T4 development (the same Basic Recursive Arithmetic framework as the companion papers). They form a small self-contained library: an *upper bound* $K(x) = O(\log x)$, a *counting bound* $\#\{x : K(x) \leq L\} \leq 3^{L+1}$, the existence of *incompressible* numbers and the unboundedness of K , an *abundance* refinement, *monotonicity*, and finally the *non-computability* of K by a Berry-program argument. Every statement is proved with no postulates and no holes; only the lower bounds use a single consistency hypothesis.

1 Setup

We work in the object theory T of Basic Recursive Arithmetic, with terms built from $\mathbf{O}$, the unary $\mathbf{s}$ and the primitive-recursive functionals; the numeral of $n \in \mathbb{N}$ is $\bar{n} = \mathbf{s}^n \mathbf{O}$, and $Thm(A)$ means “ A is a theorem of T ”. Kolmogorov complexity [3, 4] is defined relative to the universal machine of the companion Chaitin development [2].

Programs as ternary numbers. A *program* is a natural number p , read in base 3; its digits form a string fed to a fixed parser, so the enumeration of programs is the identity and “length $\leq L$ ” is the bounded condition

$$|p| \leq L \iff p < 3^{L+1}.$$

The interpreter is the object function $\mathbf{run} \in Fun_2$: $\mathbf{run}(\bar{p}, \bar{N})$ runs program p for N steps. “Program p , with fuel N , describes x ” is the provable Δ_0 equation

$$\mathbf{run}(\bar{p}, \bar{N}) = \mathbf{s} \bar{x},$$

the leading $\mathbf{s}$ separating “halted with value x ” from “not yet halted”.

The complexity predicate. For $L, x \in \mathbb{N}$ we write $K(x) \leq L$ (the relation $\preceq$) for

$$K(x) \leq L \equiv \exists p < 3^{L+1} \exists N \ Thm(\mathbf{run}(\bar{p}, \bar{N}) = \mathbf{s} \bar{x});$$

“some program of length $\leq L$ provably outputs x ”. Thus $K(x)$ is the least description length of x . Note that $K(x) \leq L$ is an *existential* over the unbounded fuel N , hence not a decidable predicate; this is why several statements below appear in negative form.

Reference semantics and consistency. Each combinator $f \in Fun_1$ has a meta value function $\mathbf{ev}(f) : \mathbb{N} \rightarrow \mathbb{N}$ (the standard primitive-recursive semantics, against which the universal machine is proved correct). The only hypothesis used by the lower bounds is simple consistency,

$$\mathbf{Con}_T \equiv \neg Thm(\mathbf{O} = \mathbf{s} \mathbf{O}).$$

2 The upper bound

Theorem 2.1 (Logarithmic upper bound). *For every $x \geq 1$ there are a program p and a fuel N with*

$$p < 3^{c+d \cdot D}, \quad \text{Thm}(\mathbf{run}(\bar{p}, \bar{N}) = \mathbf{s} \bar{x}), \quad \text{and} \quad 3^{D-1} \leq x,$$

where D is the number of base-3 digits of x and c, d are fixed constants. Consequently $K(x) \leq c + d \cdot D = O(\log_3 x)$.

The program is the base-3 Horner code of x : an honest object program whose size *nodes* is linear in D , run by the universal-machine correctness theorem; the bound $p < 3^{\text{nodes}+1}$ is the tight base-3 combinatorial estimate $2 \cdot p + 3 \leq 3^{\text{nodes}+1}$. This is the only result that needs no consistency hypothesis.

3 The counting bound

A program describes at most one value — on pain of inconsistency, since two distinct outputs of one program at a common (monotonically aligned) fuel give $\text{Thm}(\mathbf{s} \bar{x}_i = \mathbf{s} \bar{x}_j)$ with $x_i \neq x_j$, hence $\text{Thm}(\mathbf{O} = \mathbf{s} \mathbf{O})$. Counting programs then bounds the values.

Theorem 3.1 (Counting bound). *Assume Con_T . There are at most 3^{L+1} numbers of complexity $\leq L$:*

$$\#\{x : K(x) \leq L\} \leq 3^{L+1}.$$

Constructively: any sequence $x_0, x_1, \dots$ with $K(x_i) \leq L$ for all i has a repeated value among its first $3^{L+1} + 1$ entries.

The constructive form is the honest one: each value picks a describing program in the interval $\{p < 3^{L+1}\}$, and the meta pigeonhole forces two of the first $3^{L+1} + 1$ indices to share a program, hence (by determinism) to carry equal values.

4 Incompressibility and unboundedness

Theorem 4.1 (Incompressible numbers exist). *Assume Con_T . For every L it is not the case that every $i \leq 3^{L+1}$ has $K(i) \leq L$. Equivalently, some $i \leq 3^{L+1}$ satisfies $K(i) > L$.*

Corollary 4.2 (K is unbounded). *Assume Con_T . For every L it is not the case that $K(i) \leq L$ for all i .*

There are $3^{L+1} + 1$ numbers in $[0, 3^{L+1}]$ but at most 3^{L+1} of complexity $\leq L$, so one of them is incompressible above L . Because $K(x) \leq L$ is not decidable, the statement is given in the negative form “not all are compressible” rather than as a pinned witness.

Theorem 4.3 (Abundance). *Assume Con_T . No injective block of $3^{L+1} + 1$ distinct numbers can all have complexity $\leq L$.*

Taking the block to be the identity recovers Theorem 4.1; the general form shows incompressible numbers are unavoidable inside any large collection of distinct values.

Proposition 4.4 (Monotonicity). *$K(x) \leq a$ and $a \leq b$ imply $K(x) \leq b$.*

5 Non-computability of K

The library culminates in the classical observation that K itself is not computable [1, 4], by Berry’s paradox [5]. Say that a total combinator $Kf \in Fun_1$ *computes* K if it decides the complexity predicate:

$$\mathbf{ev}(Kf)(x) \leq L \iff K(x) \leq L \quad (\text{for all } L, x).$$

The two directions are soundness ($K(x) \leq L \Rightarrow \mathbf{ev}(Kf)(x) \leq L$, no over-estimate) and realisability ($\mathbf{ev}(Kf)(x) \leq L \Rightarrow K(x) \leq L$, the value is genuinely achieved).

Theorem 5.1 (Non-computability). *Assume Con_T . No $Kf \in Fun_1$ computes K .*

The proof is the Berry paradox, realised as an honest object program. Fix a hypothetical Kf . Build the fixed, μ -free combinator

$$\mathbf{berry}_{Kf} \in Fun_1, \quad \mathbf{berry}_{Kf}(L) = \text{the least } x < 3^{L+1} + 1 \text{ with } \mathbf{ev}(Kf)(x) > L,$$

a bounded search “scan the days $0, 1, \dots, 3^{L+1}$ and print the first incompressible one”. Three facts close the argument, for a suitable L .

- *The search succeeds:* writing $b_L = \mathbf{berry}_{Kf}(L)$, one has $\mathbf{ev}(Kf)(b_L) > L$. Indeed if no $x \leq 3^{L+1}$ had $\mathbf{ev}(Kf)(x) > L$, then by realisability every such x would have $K(x) \leq L$, contradicting Theorem 4.1.
- *b_L has a short description:* the program $\mathbf{berry}_{Kf} \circ \langle \text{constant } L \rangle$ runs to b_L , and its size is $nodes = \text{const} + O(\log L)$ (the constant absorbs the fixed $\mathbf{berry}_{Kf}$, the logarithm is the base-3 Horner code of the input L). By Theorem 2.1 this gives $K(b_L) \leq nodes$.
- *Exp beats linear:* since $nodes = \text{const} + O(\log L)$, there is an L with $nodes \leq L$; take $L = 3^k$ for k large, where the base-3 length of L is $k + 1$, so a constant plus a multiple of $k + 1$ stays below 3^k .

For that L we have simultaneously $K(b_L) \leq nodes \leq L$ (describable within length L) and $\mathbf{ev}(Kf)(b_L) > L$. Soundness turns the first into $\mathbf{ev}(Kf)(b_L) \leq L$, contradicting the second. $\square$

Remark 5.2. The whole development is checked in Agda under `--safe --without-K --exact-split`, with no postulates and no holes. Theorem 2.1 is consistency-free; every other result uses Con_T as its sole hypothesis, through the single fact that one program describes one value. The reference semantics $\mathbf{ev}$ is connected to the object machine by a small *derived lemma* — not a new axiom or a reflection scheme: from an object equation $f(\bar{k}) = \overline{\varphi(k)}$, the already-proved soundness of the machine, and numeral injectivity ($\overline{m} = \overline{n} \Rightarrow m = n$, itself a consequence of Con_T), one *concludes* the meta equation $\mathbf{ev}(f)(k) = \varphi(k)$. This is what lets the Berry program be reasoned about value-by-value without ever unfolding the Cantor pairing of the universal machine.

References

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- [4] M. Li and P. Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications*. Springer, 3rd edition, 2008.
- [5] B. Russell. Mathematical logic as based on the theory of types. *American Journal of Mathematics*, 30(3):222–262, 1908. (Berry’s paradox, attributed to G. G. Berry.)